

Manufacturing Sustainability Game (Preset 3)

User: \$_1 to \$_9

Initial password: ERPSIM

Adapted for Fiori and for SAP GUI
with Fiori Visual Theme Activated

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Last Update: Aug 27, 2024

CHANGE PRODUCT DESIGN Validated BOM Change (ZCS02) 1 Select the material to change by clicking Edit BOM 2 Change quantities and Save ! If received error messages, click on Continue to modify the entries	FORECAST SALES Create Planned Indep. Req. (MD61) 1 Select <i>Product group</i> and enter the following information Product group: \$\$-F 2 Continue 3 Enter your forecast quantities in the 2nd date column 4 Save	CALCULATE REQUIREMENTS MRP Run (MD01) 1 Press Enter or click Execute 2 Ignore orange warnings Press two more times on Enter 3 In the pop-up window, click Continue	SELECT VENDOR Procurement Sourcing (ZME12) 1 Click on Assign Source of Supply 2 For each <i>Material</i> , assign a vendor 3 Save	ORDER MATERIALS Create Purchase Orders (ME59N) 1 Execute ! Purchase orders are created ! If no open requisitions: <i>No suitable requisitions found</i>	RELEASE PRODUCTION Convert Planned Orders (C041) 1 Run Selection ! If no planned order: <i>Planned order could not be selected</i> 2 Select orders 3 Convert ! If conversion fails, click on ✗ to see log
PLAN STOCK TRANSFER Stock Transfer (ZMB1B) 1 In <i>Planning Mode</i> , select a <i>Push</i> or <i>Pull</i> transfer strategy 2 In <i>Scheduling</i> , enter your delivery frequency <i>If available:</i> To sell from main warehouse, in <i>Direct Sales</i> , check <i>Sales from Sloc 2</i> & specify <i>Min Qty</i> 4 Enter the amount of each product you wish to send/maintain in each region 5 Save	MAINTAIN PRICES Change Price List (VK32) 1 Open the <i>Prices</i> folder and double click on <i>Price list</i> 2 In <i>Distribution channel</i> , enter 10, 12 or 14 3 Execute 4 Modify your prices 5 Save	PLAN MARKETING BUDGET Marketing Expense Planning (ZADS) 1 Enter your daily amount of marketing expenditure for each product per area 2 Save	STANDARD INVESTMENTS Financial Postings (ZFB50) 1 Select the type of <i>standard investment</i> you wish to make 2 Enter the <i>Posting Amount</i> 3 Click on Purchase 4 Click on Post	SUSTAINABLE INVESTMENTS Financial Postings (ZFB50) 1 Select the type of <i>sustainable investment</i> you wish to make 2 Click on Purchase 3 Click on Post	PRODUCTION COST Product Cost Planning (ZCK11) Shows variable and fixed costs for each finished product Recalculates costs based on production capacity and productivity level Shows daily amounts of fixed costs (overhead, depreciation and S, G & A)
STOCK LEVELS Inventory Report (ZMB52) Shows stock levels for both finished products and raw materials Shows quantities of raw materials reserved for production	PROCUREMENT TRACKING Purchase Order Tracking (ZME2N) Shows the details/status of each purchase order Shows expected goods delivery Date	SALES AND MARKET DATA Summary Sales Report (ZVC2) Shows aggregate daily sales by product Detailed Sales Report (ZVA05) Shows sales related info such as sales revenue by region Price Market Report (ZMARKET) Shows aggregate market sales data of past 5 days	FINANCIAL STATEMENTS Financial Statements (F.01) 1 In <i>Company Code</i> , enter your <company code>* GUI step Select ALV Tree Control 2 FIORI step In <i>Statement Version</i> , enter SIM1 3 GUI step Execute FIORI step Go	MANAGE IT REPORTS Report Management (ZITM) Shows report availability and allows report(s) purchase CASH FLOW Liquidity Planning (ZFF7B) Displays an estimate of your cashflow for the coming weeks	PRODUCTION SCHEDULE Production Report (ZC00IS) Shows released production orders For each order, the time released, started and finished (or to start and finish if incomplete) If Target Qty > Conf. Qty production is still pending

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Bill of Materials

Nut		Blueberry		Strawberry		Raisin		Original		Mixed	
\$\$-F01	500g	\$\$-F02	500g	\$\$-F03	500g	\$\$-F04	500g	\$\$-F05	500g	\$\$-F06	500g
\$\$-F11	1kg	\$\$-F12	1kg	\$\$-F13	1kg	\$\$-F14	1kg	\$\$-F15	1kg	\$\$-F16	1kg
20% wheat* 30% oat* 20% nut* 1 box / 1 bag*		20% wheat* 30% oat* 20% blueberry* 1 box / 1 bag*		20% wheat* 30% oat* 20% strawberry* 1 box / 1 bag*		20% wheat* 30% oat* 20% raisins* 1 box / 1 bag*		20% wheat* 30% oat* 1 box / 1 bag*		20% wheat* 30% oat* 30% fruits & nuts** 1 box / 1 bag*	
*minimum		*minimum		*minimum		*minimum		*minimum		*minimum **requires all fruits/nut	

STORAGE CAPACITY AND COSTS

Product Type	Current Space	Daily Cost per additional 50 000 units*	Daily Carbon cost per additional 50 000 units*
Finished products	250 000 boxes	€500	2 500 (kg of CO ₂ e)/day
Raw materials	250 000 kg	€1 000	5 000 (kg of CO ₂ e)/day
Packaging (bags and boxes)	750 000 units	€100	1 500 (kg of CO ₂ e)/day

*Billed automatically

SUPPLIERS

Vendor	V01	V11	V02	V12
Lead time (days)	2-3	1-4	2-3	1-4
Delivery Cost (euros)	-	€ 1 000	-	€ 2 000
Delivery Carbon (kg of CO ₂ e)	10 000	10 000	6 000	15 000

CUSTOMERS



DC 10: Hypermarkets
Payment Time: 20 days
Approximate Market Size
€90 000 per team per week



DC 12: Grocery Chains
Payment Time: 10-20 days
Approximate Market Size
€360 000 per team per week



DC 14: Independent Grocers
Payment Time: 1-20 days
Approximate Market Size
€135 000 per team per week

TRANSPORTATION AND CARBON FEES

Movement type	Cost (€)	Carbon (kg)
Main Warehouse to Regions	500	750
Regions to Customers	-	200
Main WH to Customers (per unit)	0.05	0.25

VARIABLE CARBON TAX

Initial Price (€/kg of CO ₂ e)	0.20
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SUSTAINABLE INVESTMENTS

Type	Cost (€)	Carbon (kg of CO ₂ e)	Reduction (%)	Max. Reduction (%)
Freight Fleet Improvement	10 000	2 000	15	45
Sustainable Manufacturing	10 000	2 000	15	45

FIXED COSTS (€ paid each 5 days)*

Labor	20 000
Manufacturing overhead	15 000
S, G & A	40 000
Depreciation (Building)	1 250
Depreciation (Equipment)	50 000

*Billed automatically

DAILY OVERHEAD CARBON

Purchased Energy (kg of CO ₂ e)	500
Other Overhead (kg of CO ₂ e)	400

PRODUCTION CONSTRAINTS

Capacity (units/day)	24 000
Additional Capacity Cost (€ per 1 000 units)	1 000 000**
Additional Capacity Carbon Emission (kg per 1 000 units)	1 000
Production Carbon Emission	0.30 kg per box
Setup Carbon Emission	50 kg per hour
Minimum/Maximum Lot Size	16 000/48 000

**Investing in additional capacity will increase equipment depreciation costs

SETUP TIME REDUCTION

Setup time (hrs)	Cost (€)	Carbon (kg)
8	-	-
7	50 000	100
6	125 000	250
5	250 000	500
4	500 000	1 000
3	1 250 000	2 500